The Rogowski Coil Principles and Applications: A Review

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***Abstract*— The Rogowski coil is an old device for current measurement. It has been modified and improved over a century and is still being studied for new applications. The Rogowski coil has various advantages over conventional magnetic current transformers (CTs). Not only can it be used instead of CT, but also it has various utilizations in other fields. This paper provides a brief review on different aspects of the Rogowski coil and its advancement procedure, during last decades. In this paper, the history of the coil is brifely reviewed and its bases and applications are discussed. The Rogowski coil is analyzed from different points of view, including different integration techniques in the output stage, models for the Rogowski coil, experimental methods for parameter measurement in models, and method for determining the damping resistor. Finally, a brief review over different applications of the coil ends this paper.**

***Index Terms*— High-speed current measurement, lumped and distributed model, review, Rogowski coil, specific applications.**

I. Introduction

THE Rogowski coil is an air-core coil, which measures both alternating and high speed impulse currents, based on Ampere's and Faraday's law [1]. It was named after a German physician Walter Rogowski [2]. Firstly, this type of transducer proposed in 1887, when A. P. Chattock from Bristol University was working on the better types of dynamos [3]. He used a long coil on a plastic rod for measuring the magnetic reluctance. He attached the two ends of the coil to each other, and calibrated the device based on Ampere's law [4]. In 1912, W. Rogowski and W. Steinhaus used Chattock's technique for magnetic potential measuring. In this manner, They performed various tests to ensure the validity of the coil measurements [4], [5].

The main limitation regarding the Rogowski coil applications was about diminutive output in measuring low amplitude currents [6]. In first stages, the coil usage was limited for measuring the high amplitude currents, with high variation rate, due to the fact that, the output of the coil is proportional to the derivative of the current [3], [7]. Nowadays, the Rogowski coils are able to measure low level currents, thanks to electronic devices [8]. This coil does not have ferromagnetic core, therefore, it has a linear characteristic [9]. Linear characteristic together with accurate electronic devices, make it possible to measure currents, form milli-amperes to mega-amperes, using Rogowski coil. Furthermore, the low cost of this device, comparing to the other measurement methods, makes the Rogowski coil an appropriate gadget for measuring high amplitude transient current [2], [10]–[12].

The output of the Rogowski coil was insufficient in conventional measuring methods, which was the main limit in past decades. However, nowadays by developments of microprocessor-based measurement devices, Rogowski coils are more suitable for various applications. The nature of Rogowski coil measuring method, which measures the current derivative, limits its usage in measuring DC currents [10]. The Rogowski coil has some significant features, which attract attentions in recent years. Some of the main advantages are as follows [13]:

- Enduring large overloads without damage;
- Measuring currents in an extensive range, without saturation;
- Easy to use, due to the flexibility and light weight;
- Low cost;
- Nonintrusive nature (drawing no power from the main circuit);
- Wide bandwidth, in a range of 0.1 Hz to 1 GHz;
- Excellent transient response;
- Safety (isolated from the main circuit, electrically).

This paper is a review on this fascinating device. The rest of this paper is organized as follows. In Section II, the basics of Rogowski coil are presented. Various ways of integration in its output is discussed in Section III. The paper proceeds with introducing different models of Rogowski coil and their features in section IV. Section V, reviews some applications of Rogowski coil and finally, Section VI provides the conclusion. This paper is a very good reference for the engineers who want to recognize and utilize Rogowski coil and need a fast guide.

II. Basics of Rogowski Coil

A typical Rogowski coil consists of toroidal form windings, encircling the current path, as shown in figure 1 [14]. Referring the Ampere's law, the relation between the current flowing through the Rogowski coil and the intuitional magnetic, along the axis of the torus, is as follows:

$$I(t) = \frac{1}{\mu_0} \oint B(\vec{t}) \cdot \vec{ds}, \tag{1}$$

where s is the distance along the torus. Magnetic field B variations, induces voltage in the windings. Considering the

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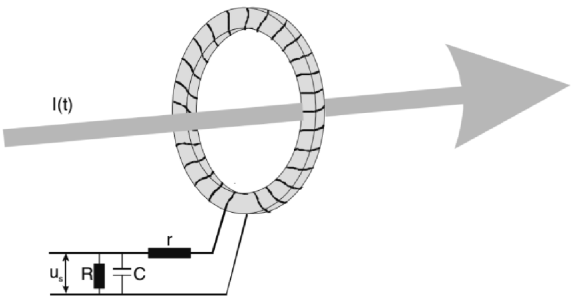


Fig. 1. A sample Rogowski coil with an RC integrator [15].

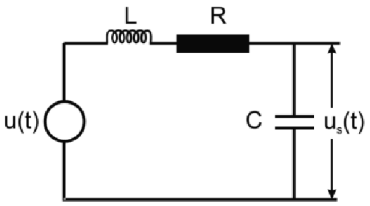


Fig. 2. Equivalent circuit of a single-turn winding with an RC integrator [15].

constant cross-section for windings and constant turns number per length unit, a unique relation between the induced voltage, in the terminal of the Rogowski coil $u(t)$ and the flowing current $I(t)$ can be written based on Faraday's law [15]:

$$u(t) = \frac{d\phi}{dt} = \int \vec{B(t)} \cdot \vec{dA} = \frac{A}{s}\mu_0 \dot{I(t)}, \tag{2}$$

where A is the windings cross-section and s is the number of turns per length unit.

Formula (2) is independent of the current distribution. For deriving the current value from induced voltage $u(t)$, an integrator block is necessary at the coil output terminal. A simple RC circuit can acts as integrator block, as shown in figure 1. Figure 2 shows a simple RC circuit which is configured as an integrator. The self-inductance of each winding is modeled as L in figure 2. R and C are the resistor and capacitor of integrator circuit, respectively. If the current i flows in this circuit one can write:

$$u(t) = \frac{d\phi}{dt} = L\frac{di}{dt} + Ri + \frac{1}{C}\int_0^t i dt', \tag{3}$$

ω is assumed as the highest frequency in the Fourier transform spectrum of the source current. By setting $L\omega \ll R$ then the (3) can be rewritten as follows:

$$u(t) = \frac{d\phi}{dt} = Ri + \frac{1}{C}\int_0^t i dt'. \tag{4}$$

Considering the measurement time, t, much less than RC value, then the below formula can be derived for $u_s(t)$ [15]:

$$u(t) = Ri(t); u_s(t) = \frac{1}{C}\int_0^t i dt' = \frac{1}{RC}\int_0^t u(t')dt' = \frac{A\mu_0}{sRC} I(t) = \mu_0 \frac{NA}{S_m}\frac{1}{RC} I(t), \tag{5}$$

where N is the number of windings, A is the cross-section of the coil, and S_m is the mean length of the coil. It should be noted that, $L\omega \ll R$ is the necessary condition in (4) and (5). In measuring pulse shape currents, this condition is not fulfilled, due to the high frequency components in the current wave spectrum. In such a condition, where the sum of coil resistance, r, and external resistor, R, is less than $L\omega$, and $R \ll 1/\omega C$, as well, then (3) can be written as follows:

$$\frac{d\phi}{dt} \approx L\frac{di}{dt} \Rightarrow i = \frac{\phi}{L} = \frac{\mu_0 NAI/s}{\mu_0 N^2 AI/s} = \frac{I}{N}$$
$$\Rightarrow u_R(t) = iR = \frac{R}{N} I. \tag{6}$$

In this case, the output of the coil is proportional to the flowing current. Thus, it has linear characteristic, like a conventional CT [10], [15]. According to the equations, the lowest measurable frequency is determined from $r + R \ll L\omega$ condition and the highest measurable frequency is depend on the LC resonance frequency of the coil itself. Therefore, in high frequency measurements, the Rogowski coil has an intrinsic integrator and does not need external integration block [3].

The other problem relates to the capacitive coupling between coil windings and the casing. This coupling causes error in the output signal and should be omitted for increasing the accuracy, in most of the cases. This coupling effect is a source of considerable error in high frequency measurements. In this situation, the coil acts like a transmission line and the induced voltage of different spots reach the coil terminal with different delay time, which result in pulse distortion. Moreover, uneven excitation of different windings can cause a remarkable oscillation in the output of the coil. Consequently, the output has direct relation with the current distribution in the torus.

Regardless of the mentioned problems, if the Rogowski coil is in closed loop structure, the coil can have any shape. This posture refers to the ampere's law essence which says, the closed loop integral of the magnetic field around a current path is equal to the current. In other words, the trajectory of flowing current has no effect on the output. This fact results in a outstanding feature of the Rogowski coil as flexible device for current measurement. Hence, the Rogowski coil is the best and only choice in most cases, where other devices such as CTs do not have the desired flexibility [16].

Despite simple structure of Rogowski coil, in utilizing stage, it needs some special considerations. One of the problems is related to the large magnetic flux between two ends of the coil. For decreasing this flux and increasing the immunity of the coil versus stray fields, the Rogowski coil should include two electric paths which are connected to each other in reverse direction [13]. Windings can be either both of the paths or just one of them. When both of the paths are in winding form, they should be wound reversely, in order to intensify the output [17], [18]. While, when one path is winding and the other is a simple wire, the second wire can be returned through inside of the first wound path. In this case, the coil can be separated from one end. Easy separation structure is suitable for measuring in cases where the primary path cannot be opened. This configuration is shown in figure 4(a).

III. Output Integration in the Rogowski Coil

As it stated earlier, in normal operation mode, an integrator circuit is required at the output of the Rogowski coil.

One of the intrinsic advantages of this integrator is noise reduction because of low-pass frequency characteristic, which is important in high precision measurements [7]. There are various approaches for implementing this integrator. The RC integrator has been discussed in the previous section. The self-integrating mode of Rogowski coil in high frequency measurements presented, as well. Another group of methods for designing integration circuit is based on using operational amplifier (OpAmp). As mentioned, RC-base integrator is appropriate for high frequencies ($> 100MHz$), while, OpAmp-based integration method is useful in low frequency ($< 100MHz$) measurements [19].

Another applicable integrating method is based on using a microprocessor based devices with an analog to digital converter [20]. In these methods, a signal processing software operates as an integrator. In signal processing based approaches, additional block like digital filters can result in better outputs. Also, adaptive algorithms are useful for precise waveform reconstruction [20], [21] which is important for wideband measurements. This method is appropriate for mid-frequencies.

When there is a low impedance resistor, in series with coil, the Rogowski coil acts as a self-integrator [10]. However, in this situation the gain of the coil is very low. One option to intensify the coil sensitivity is using multiple coils in series. In fact, if the mentioned low impedance resistor is much less than $L\omega$, then the coil works in linear mode. In the other hand, if the termination resistor is much more than $L\omega$, then the coil works in derivative mode. The self-integrating mode is suitable for impulsive current measurement [2].

The perfect integrator should satisfy some requirements. Not only it should have a wide frequency bandwidth, but also, it should have a time constant, which is multiple times greater than the main circuit time constant [22]. The RC integrator faces with variety of rigid limitations, which are hard to satisfy. For example, in power frequency measurement (50–60 Hz), both R and C have high values. Thus, the integrator does not have a suitable frequency performance, as a result of stray capacitances, parallel with resistor and the capacitor dissipation. Furthermore, the output to input ratio is very small which results in low sensitive measurements. Because of these problems and amplification necessity, the integrator with OpAmp circuits are preferred in power frequency measurement. Some researches have suggested compound integrators with wider bandwidths [23].

The basic OpAmp integration circuit is shown in figure 3(a), however, there are other advanced integration OpAmp circuits which their design is based on this structure [24], [25]. Ignoring the R_1, the gain of integrator would be $(1+G)R_2C$, in which G is the open-loop gain of OpAmp. In this case, any noise in the input enforce the output into the saturation condition. Adding R_1 to the basic circuit, decreases the DC gain of integrator down to R_1/R_2. The frequency characteristic of this circuit is shown in figure 3 (b). In figure 3 (b), the f_1 and f_0 are as follows:

$$f_0 = \frac{1}{2\pi R_1 C_1}, \qquad f_1 = \frac{1}{2\pi R_2 C_1}. \tag{7}$$

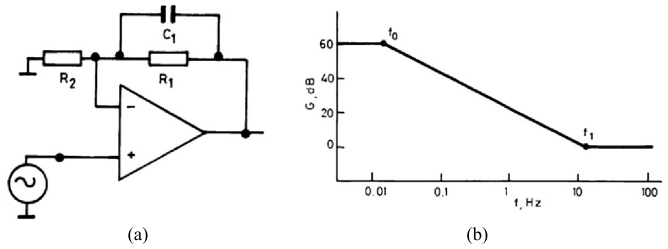


Fig. 3. OpAmp integrator. (a) Basic OpAmp circuit. (b) Frequency characteristic [22].

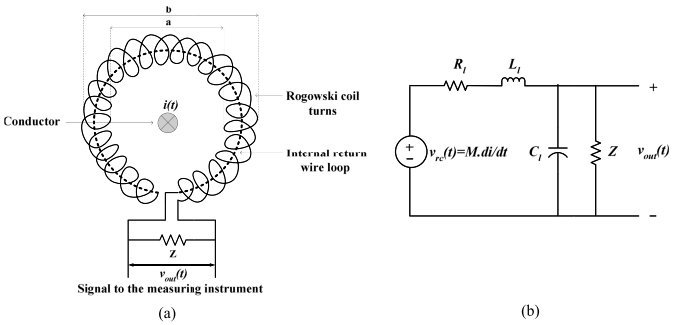


Fig. 4. (a) A Rogowski coil structure. (b) Lumped model of Rogowski coil [26], [27].

The circuit acts as an integrator in bandwidth between f_0 and f_1. Reference [22] proposed a combinational method which does the integration process in three different bandwidth, with three different methods. The integration in low frequencies is done using an OpAmp circuit; in mid-frequencies an RC circuit is utilized; and in high frequencies the coil utilized in the self-integration mode. Therefore, a wider frequency bandwidth is supported using the proposed scheme.

IV. Rogowski Coil Modeling

A. Lumped Model

In this subsection, the lumped model of Rogowski coil is reviewed. The lumped model of Rogowski coil is acceptable for low frequency analysis [25]. It includes a resistor, an inductor, and a capacitor. Figure 4 depicts this model. The parameters in this figure are computable as follows [23], [26]:

$$R_l = \rho_c \frac{l_w}{\pi d^2}, \tag{8}$$

$$L_l = \frac{\mu_0 N^2 d_{rc}}{2\pi} log\frac{b}{a}, \tag{9}$$

$$C_l = \frac{4\pi^2 \varepsilon_0 (b+a)}{log(\frac{b+a}{b-a})}, \tag{10}$$

where ρ_c is the electrical resistivity of the coil wire, l_w is the length of coil wire, d is the radius of this wire, d_{rc} is the diameter of each loop in the coil, N is the number of turns, and a, b are the internal and external radius of the coil, respectively. Z is modeled as an external impedance in the coil terminals, includes damping resistor, coupling cable capacitor and measurement instrument impedance.

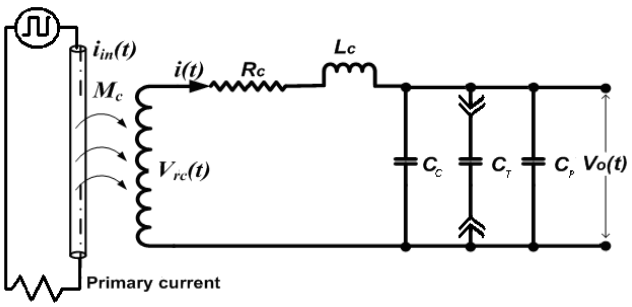


Fig. 5. Experimentla setup for measuring parameters of the Rogowski coil [19].

The equations are credible in the ideal mode, while, in the construction, there are some imperfections; *e.g.* distribution of turns along the coil is not even, the return path is not in the center of windings, and the cross-section is not circle, ideally and it turns to an ellipse due to bending. Because of these imperfections, the above formulas are not accurate [19]. For high-frequency applications, exact parameter determination is important, since the bandwidth and choosing the damping resistor in the terminal is related to these parameters values. Furthermore, issues like proximity and skin effect cause the erratic distribution of current in the coil, hence, some parasitic inductors and capacitors should be considered in the model [19]. Defining these parameters with computational methods is not accurate enough, while it needs remarkable time owing to large computational size. Hence, it is better to measure these parameters after the coil construction. However, reference [11], [28] has proposed a computational method based on PEEC (Partial Element Equivalent Circuit), which is fast and the results has a great accordance with measured parameters. Measuring the lumped model parameters is discussed in the next subsection.

1) Parameter Measurement: There are different methods for measuring the lumped model parameters of the Rogowski coil. One of them is reviewed here. The measurement circuit is shown in figure 5. C_p is the capacitor of measurement unit and C_T is an external capacitor added to the system. C_l and L_l are the parameters, which should be determined. In this method, the first resonance frequency of the system should be measured. This frequency is measurable using two different techniques [19]. In first method, a pulse using a pulse generator (*e.g.* the partial discharge calibration instrument) is applied to the primary circuit. Afterwards, the output voltage is recorded and the first frequency in voltage waveform can be determined using the Fourier spectrum. Due to the low coupling between the primary and the secondary sides, the measurement setup should be precise enough.

The second technique is based on using a signal generator in the primary and controlling the secondary voltage, simultaneity. As the frequency of generator is increased gradually, the first frequency, in which the output voltage becomes maximum, is the first resonance frequency of the system.

Determining the first resonance frequency of the system is useful in lumped model parameter measurement. At first, a capacitor C_{T1} is added to the system and the resonance frequency is determined. Then, another capacitor C_{T2} is replaced and the new resonance frequency is determined, as well. For resonance frequency, the formula $f_{res} = 1/2\pi\sqrt{L_l C}$ can be used considering $C = C_l + C_T + C_p$. Since the resonance frequency is defined with two different C_T one can write:

$$\frac{f_{res1}}{f_{res2}} = \sqrt{\frac{C_l + C_p + C_{T1}}{C_l + C_p + C_{T2}}}. \tag{11}$$

The resonance frequencies, C_{T1} and C_{T2} are known, therefore, $C_l + C_p$ value is calculable. By knowing the resonance frequency and capacitors values, the inductance of the coil is calculable. When C_p and C_l are required separately, the experiment can be repeated with two different C_p, *e.g.* two different probes, thus with two equations C_p and C_l can be determined separately [19]. The model parameters can also be determined using impedance analyzers [2]. Specifically, this method is suitable for defining distributed model validity.

2) Damping Resistor Determination: The lumped model of the Rogowski coil is valid, in measurements below the first resonance frequency of the coil. Based on the lumped model the transfer function of the coil can be driven [28]:

$$H_{coil} = \frac{V_{coil}}{i} = \frac{s.M}{L_l C_l.s^2 + (R_l C_l + \dfrac{L_l}{Z}).s + 1 + \dfrac{R_l}{Z}}. \tag{12}$$

The resistance of the coil is negligible. Therefore, the transfer function can be rewritten as follows:

$$H_{coil} = \frac{V_{coil}}{i} = \frac{s.M}{L_l C_l.s^2 + \dfrac{L_l}{Z}.s + 1}. \tag{13}$$

The transfer function shows that, the output voltage is proportional to the derivative of the current. Denominator is a second order polynomial function. Hence, without damping resistor, an overshoot occurs near the resonance frequency $\omega_0 = 1/\sqrt{L_l C_l}$. This condition cause no problem for power frequency measurements, however, if the primary current contains frequencies components, near the coil resonance frequency, the overshoot may affect the operation of the system. For example, in protection functions, high-frequency components of the current, may cause overshoot, which result in the relay maloperation. This phenomenon is noticeable in protecting the power electronic instruments [29], owing to high frequency transients during switching times [28]. Consequently, a damping resistor is necessary in most of the cases.

In case of using an integrator block in the output stage, the simplified transfer function for the integrator output can be written in the following form:

$$H(s) = K.\frac{\omega_0^2}{s^2 + 2\zeta\omega_0.s + \omega_0^2}. \tag{14}$$

Thus, the resistor value derived using following formula [11], [25]:

$$R_d = \frac{1}{2\zeta}\sqrt{\frac{L_l}{C_l}} = \frac{1}{\zeta}\pi L_l f_{res}, \tag{15}$$

where $\zeta = 1$ is a good value for resistor determination [30]. This resistor should be an inductance-free resistor, otherwise it origins new oscillations. Damping resistance value is

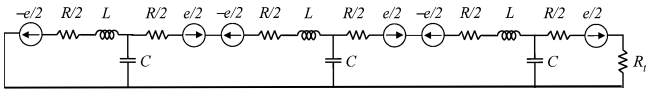


Fig. 6. Distributed model of the Rogowski coil [13].

calculable, based on root locus analysis or Bode diagram, accurately [2]. However (15) accuracy is sufficient for not-highly precision measurement.

B. Distributed Model

As discussed in previous part, the lumped model is not valid in high-frequency measurements, thus it should be replaced with a distributed model. One of the distributed model of the Rogowski coil is based on transmission line model theory [31]. The model based on transmission line theory is used for achieving following purposes:

- Optimization of the coil parameters, particularly the shunt capacitance in order to reach self-integration mode;
- Determining the effect of termination resistance, on the output voltage of the coil;
- Decreasing the adverse effect of reflection by declining the stray capacitances.

In this model, the electrical distance of the coil is separated into n divisions. Each division contains series inductance, $L = L_l/n$, modeling the inductance of the division, a shunt capacitor $C = C_l/n$, modeling the stray capacitance of the division, a resistor $R = R_l/n$, and a voltage source $e = u_t/n$ modeling the front and backward travelling wave [23]. The distributed model of the Rogowski coil is shown in figure 6.

The coil sensitivity, travelling time of the wave in the coil, front time of the wave, and pulse width of the current are most important factors that should be considered in measuring currents with severe variations. Moreover, for achieving self-integrating mode, the termination resistor should be considered equal to the least possible value. Theoretically, the best state for self-integrating mode is short circuit condition of the terminal. However, it is not practically true, since for measuring output voltage, the resistance existence is necessary.

In manner of obtaining best transient response, the shunt and inter-turn capacitances should be minimized [7]. In fact, the shunt capacitor is the stray capacitance between the coil windings and the primary conductor, while the inter-turn capacitor is the capacitance of coil loops between each other. Various techniques are suggested for decreasing these capacitances. One of these techniques is based on putting a shield around the Rogowski coil, to decline the shunt capacitors [13], [32]. This shield can also be used as a return path. Changing the space between the coil windings and the shield sheet leads to different shunt capacitors. It is necessary to consider an overall slot through the sheet; otherwise the coil cannot work properly due to magnetic shielding of the sheet.

The inter-turn capacitance is related to the windings density. As the winding density increase, the inter-turn capacitance become higher, as well. Thus, larger space between windings result in less capacitance [32]. Another technique for decreasing this capacitance is based on inserting an extra winding between the main coil windings. In the manner of decreasing

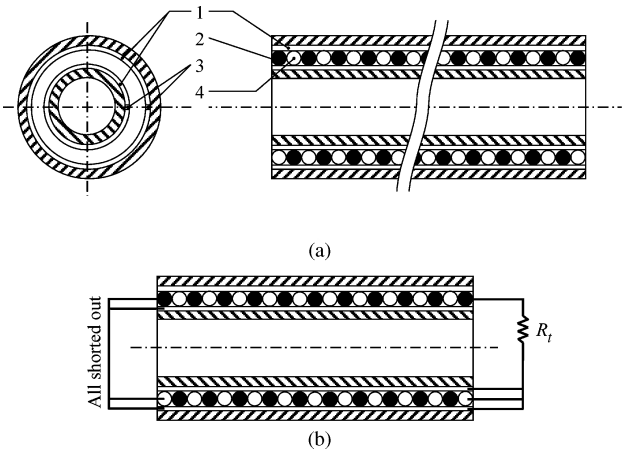


Fig. 7. (a) Details of a self-inegrating Rogowski coil for measuring currents with intensive variation: 1) inside and outside aluminum shield (return path), 2) insulated conductor of the coil, 3) longitudinal slots on the return path, and 4) shielding conductor for inter-turn capacitance. (b) The connections at both ends of the coil [13].

the overall coupling capacitance, the extra winding should be connected to the shield. This method causes considerable reduction in the inter-turn capacitance value [13]. Figure 7 shows some of these techniques.

Increasing the number of windings loops in the coil, intensifies the sensitivity, however, raises the coil inductance and decreases resonance frequency, as well. As mentioned before, in applications which need a wider bandwidth, the resonance frequency should be higher. In fact, specifying the number of loops is a trade-off between coil sensitivity and its usable bandwidth [19], [31]. With less number of turns, it is possible to identify the coil parameter using analytical equation [1], [31], [33]. Termination resistor value has impact on the coil output, in high frequencies. The best measurement performance in high frequencies can be derived from shunt resistors, but they are rather expensive. However, optimal setting of the termination resistance, which cost much lower than shunt resistor, results in appropriate output, very similar to the shunt resistor [16], [34]. More details about setting the termination resistance value, in high frequency applications, is available in [13], [34], and [32].

V. Applications of the Rogowski Coil

Not only, the Rogowski coil is useful in conventional applications, but also, there are other cases where the Rogowski coil is the best choice [7]. Because of non-geometric structure and capability in measuring high amplitude current and etc., it has various applications in testing laboratories, industrial measurement instruments, and unconventional measuring systems [29]. In following, a brief overview of the Rogowski coil applications is presented.

In the power system protection, conventional CTs are being used for decades. These CTs were necessary for electromagnetic relays operation, due to the large power demand of these relays for proper operation. Nowadays, the relays are numeric and work with low power signals properly. As a consequence, the Rogowski coil can be used instead of CTs for fault detection in transmission lines and electrical machines [35].

Using Rogowski coil leads to fast response of the relays and flexibility of protection schemes [8], [36]. This matter has also opened new gates to the protection schemes fields [35]. For example, the Rogowski coil owing to wide frequency bandwidth can measure the travelling wave variation, which enables using traveling wave based protection schemes [3]. Travelling wave based algorithms is useful in fault location methods [23]. It is also possible to add the coil on distribution overhead lines without primary separation for high-impedance fault detection [26].

Some of the CT problems, such as CT saturation, which lead to maloperations of the relays, are not the matter of concern in the Rogowski coil. Recently, numerous researches are focused on using the Rogowski coil instead of CTs in transmission and distribution substations [3], [7], [23]. The Rogowski coil is useful as an electronic current transducer in smart substations for protection and monitoring applications, as well [24]. Owing to light weight and high precision [9], [33] the coil can be used for on-site calibration of CTs, in substations [29], [37], [38]. Furthermore, the coil have been used vastly for current measurement in Gas Insulated Substations (GIS) by great industries [7].

The Rogowski coil can be used in differential protection of furnace transformers. These transformers conduct tens of kilo amps in the secondary for electrowinning applications. The conventional protection of these transformers utilizes the measurements of primary side. However, the differential protection is the best option for transformer protection. Since, CTs which are able to measure high level currents are expensive, the differential scheme is not economical. The Rogowski coil can measure high amplitude currents, without problems such as saturation. Consequently, differential protection schemes are executable for furnace transformer [17]. Referring to the same considerations, the Rogowski coil is the best choice for high current measurement in high power laboratories [37].

As mentioned, the Rogowski coil measures the current derivative [39]. Therefore, it is useful in triggering applications. For example, in the synthetic test circuit of high voltage circuit breakers, current is applied to the breaker from one circuit, and the transient recovery voltage is applied to the breaker after zero-crossing of the current by another circuit. In the first zero-crossing, the current should be interrupted. The moment of current interruption and voltage creation should be synchronized as much as possible. For this purpose the moment of current initiation is measured and the voltage is applied after a certain time. When the current flows, the current derivative is remarkable. Thus, the Rogowski coil can sense this variation and prepare a voltage in its terminals for rest of the setup [40]. In the same way the Rogowski coil can be used for protection of power electronic switches against high di/dt or short circuit current [11], [41]. Detection of plasma ignition is possible, with the same method, using the Rogowski coil [42]. The coil has other applications in circuit breaker area. For decades, measurement of pre-arc and post-arc current in a circuit breaker was impossible, due to the fact that the first one is very large, while the latter is quite small. The point is that the derivative of current during arc quenching in a breaker is large and then the Rogowski coil has provided a possibility to measure these currents [7].

Moreover than wide bandwidth, capability of measuring currents with high amplitude, makes the Rogowski coil one of the best options for measuring large, sharp currents [25]. These currents are used vastly in pulsed power application, and the Rogowski coil is one of the best measurement devices in plasma and pulsed power technology [28], [42]. It is also a good choice as a measurement device in tests, dealing with impulse currents [2], [12], [34]. It is used in high-tech TOKAMAK fusion devices for normal and eddy current measurements [43], [44] and in plasma setups, where the current does not flow in a conductor but in an area in the air [42].

Another application for the Rogowski coil is real-time Partial Discharge (PD) measurement [39], [45]. The PD is a high frequency phenomenon and its measurement needs a device with wide bandwidth [27], [32]. Detection of defects in high voltage components and disconnecting them, from grid before a serious damage can save money and time for power grid. One of the possible methods for detection of defects in high voltage components is PD measurement of instruments. Generally, preceding a breakdown in high voltage apparatuses, the rate of partial discharge rises [46]. Therefore, by monitoring the partial discharge, the apparatus status is determinable. The Rogowski coil can be used for online and offline PD measurement, due to wide bandwidth [19], [20], [32]. In some cases, like in power cables, PD detection based on voltage signals is quite difficult, because of stray capacitances. In such situations, the Rogowski coil is a proper choice for PD measurement, based on measured current signals [47], [48].

Recently a Rogowski coil is constructed using PCB (Printed Circuit Board) [49]. This type of coil is very accurate, due to precise sizing of windings. In this coil, top layer becomes one edge of winding wire and the bottom edge plays as the other wire. The two other edges lay inside the board connecting the top and bottom layer to each other via holes [50]. This type of Rogowski coil is used for partial discharge and impulse current measurement [11]. This type is a good choice for precision measurement applications with digital wiring implementation and extensive automated production. This type of Rogowski coil does not suffer from uneven density or reshaped windings [2] and its precise characteristic is derived analytically in the literature [51]. Moreover PCB coil can be designed in a way to have high temperature stability which is important for industrial applications [7].

In applications where there is no normal geometrical structure, the Rogowski coil is one of the best options, due to its flexibility [16]. For instance, one Rogowski coil can be placed around the tower, for measuring the impulse current distribution in a wind turbine tower [12]. Measuring the leakage current of a system, which placed on the earth, can be done by placing big Rogowski coils around all of the legs of that component. Using this method, the grounding resistance of overhead line towers can be measured, individually [52]. The coil can also be placed around high-voltage bushings, for on-line monitoring [3]. It is also useful for eddy current measurements in non-geometric structures [44].

The Rogowski coil have intrinsic insulation from main circuit. Using optical link between the coil and the measurement system has strengthened the coil for using in high voltage applications [53], [54]. This area can be developed vastly in the future. Moreover than this field, increasing coil bandwidth, which is important in high-frequency and precise measurement, is of great interest these days and needs further studies [25].

VI. Conclusion

In this paper, different aspects of the Rogowski coil were presented. First, the history of the coil was brought up. Then, the basics of Rogowski coil is discussed and different methods about output integration block is reviewed. Afterwards, two models of the Rogowski coil, lumped and distributed, were discussed. Experimental methods for determining the lumped model parameters were reviewed and a simple approach for choosing appropriate damping resistor, in the terminal of the coil, was presented. At last, various applications of the Rogowski coil was reviewed and the requirement for each application was discussed. Despite the fact, that this coil is invented decades ago, there are different fields, in which the research about the Rogowski coil is still continuing.

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